VITAMIN DEFICIENCY DISEASE PREDICTION USING MACHINE LEARNING

1. INTRODUCTION

Vitamin deficiencies occur when the body does not get enough essential nutrients such as Vitamin A, Vitamin C, Vitamin D, Iron, or Folate. These deficiencies can lead to diseases like Anemia, Night Blindness, Scurvy, and Rickets/Osteomalacia.

Lifestyle factors such as diet, smoking, alcohol consumption, exercise, and sun exposure can strongly influence these conditions. Traditional diagnosis may require multiple tests, which can delay early detection.

This project aims to build a machine learning model that predicts vitamin deficiency diseases using health, lifestyle, and nutritional data. By using data pre-processing, feature selection, and model comparison, the project identifies the best model for accurate disease prediction.

2. PROBLEM STATEMENT

Vitamin deficiency diseases are common and may be difficult to identify early because symptoms can overlap across multiple conditions. There is a need for a predictive system that can analyze patient data and accurately identify possible vitamin deficiency diseases.

This project focuses on developing a machine learning-based system that predicts diseases using patient symptoms, nutrition levels, and lifestyle factors.

Machine Learning → Supervised → Classification

3. OBJECTIVES

- To pre-process and clean the vitamin deficiency dataset
- To perform Exploratory Data Analysis (EDA) for pattern identification
- To apply feature selection for choosing important features
- To train and compare multiple machine learning models
- To evaluate models using Accuracy and F1-Score
- To select the best-performing model for final prediction
- To deploy the final model for predicting diseases from raw user input

4. DATASET DESCRIPTION

The dataset used in this project is the Vitamin Deficiency Disease Dataset. It contains patient health information related to nutrition, lifestyle, symptoms, and clinical factors.

The dataset includes:

- Demographic details
- Diet type
- Smoking and alcohol habits
- Exercise level
- Sun exposure
- Vitamin and nutrient levels
- Deficiency-related symptoms

These features help in understanding the health condition of an individual and predicting possible vitamin deficiency diseases.

Target Variable:

The target column is *disease_diagnosis*, which classifies individuals into:

- Anemia
- Healthy
- Night Blindness
- Rickets/Osteomalacia
- Scurvy

This dataset was used for data pre-processing, exploratory data analysis, feature selection, model building, and final prediction.

5. DATA PREPROCESSING

Data pre-processing was performed to clean and prepare the dataset for machine learning model development.

Steps involved in pre-processing:

- Removed unnecessary columns such as *has_multiple_deficiencies* and *symptoms_list*
- Checked for missing values
- Applied One-Hot Encoding for categorical columns like *gender, smoking status, exercise level, and diet type*
- Applied Label Encoding for ordinal features such as *alcohol consumption, sun exposure, income level, and latitude region*
- Converted Boolean values into integer format
- Encoded the target variable *disease_diagnosis* into numerical classes

These pre-processing steps transformed the raw dataset into a machine-readable format, making it suitable for feature selection and model training.

6. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the dataset, identify important patterns, and study the relationship between lifestyle, nutritional factors, symptoms, and disease diagnosis.

EDA focused on:

- Disease distribution across the dataset
- Comparison of nutrient levels between healthy and deficient individuals
- Common symptoms associated with each disease
- Impact of diet type on disease occurrence
- Relationship between sun exposure and Vitamin D levels
- Effect of alcohol consumption and smoking habits on health indicators
- Gender and income-based disease distribution

Various visualizations such as count plots, boxplots, heatmaps, and violin plots were used to identify meaningful insights and trends in the dataset. This analysis helped in better understanding the factors influencing vitamin deficiency diseases.

7. FEATURE SELECTION

Feature selection was performed to identify the most important features that contribute to disease prediction and to reduce unnecessary data complexity.

Method Used:

SelectKBest with the **Chi-Square (chi2)** statistical test was applied.

Process:

- The dataset was first divided into training and testing sets
- Select Best was fitted only on the training data to avoid data leakage
- The top 10 most relevant features were selected based on their importance scores
- The same selected features were then applied to the test data

This step helped improve model efficiency, reduce dimensionality, and focus on the most significant predictors for vitamin deficiency disease classification.

8. MODEL CREATION

Multiple supervised machine learning classification models were trained and evaluated to identify the best model for vitamin deficiency disease prediction.

Models Used:

- Random Forest
- Decision Tree
- Logistic Regression
- Support Vector Machine (SVM)
- Naive Bayes

Evaluation Metrics:

- Accuracy
- F1-Score

Each model was trained using the selected features after pre-processing and scaling. GridSearchCV was used for hyper parameter tuning in applicable models to improve performance.

9. FINAL MODEL SELECTION

Based on model comparison results, **Random Forest Classifier** was selected as the final model for this project.

Reason for Selection:

- Highest Accuracy
- Best F1-Score
- Better overall classification performance
- Strong handling of multiple disease classes
- Reliable prediction compared to other models

Best Parameters:

- n_estimators = 100
- criterion = entropy
- max_depth = None

The final model was trained using the selected top features and scaled data. This model was then saved for deployment.

10. PERFORMANCE EVALUATION

The performance of the project was evaluated in two stages: feature engineering method comparison and machine learning model comparison.

10.1. Feature Selection and Dimensionality Reduction Evaluation

The proposed system was evaluated by comparing different feature engineering techniques and supervised machine learning models to identify the best-performing approach for vitamin deficiency disease prediction.

Table 1: Performance Comparison of Feature Selection and Dimensionality Reduction Techniques

Method	Technique type	Components/k	Accuracy	F1 Score
SelectKBest (Chi-square)	Feature Selection	K=10	0.935	0.936
RFE	Feature Selection	K=10	0.920	0.920
PCA	Dimensionality Reduction	10	0.749	0.735
KPCA	Dimensionality Reduction	10	0.738	0.729
LDA	Dimensionality Reduction	2	0.660	0.639

Table 1 shows that SelectKBest (Chi-Square) achieved the highest Accuracy and F1-Score, making it the most effective feature engineering technique for this dataset. Feature selection methods outperformed dimensionality reduction techniques in overall classification performance.

10.2. Machine Learning Model Evaluation

After selecting the best feature method, multiple supervised classification models were trained and compared.

Table 2: Performance Comparison of Supervised Classification Models

Model	Accuracy	F1-Score
Random Forest	0.93	0.92
Decision Tree	0.924	0.923
SVM	0.862	0.861
Logistic Regression	0.823	0.822
Naïve Bayes	0.717	0.71

Table 2 indicates that Random Forest achieved the best overall performance among all supervised classification models. Based on its strong Accuracy, F1-Score, and reliable multi-class prediction capability, Random Forest was selected as the final model for deployment.

Overall, the evaluation confirms that SelectKBest with Random Forest provided the most accurate and reliable solution for vitamin deficiency disease prediction.

11. DEPLOYMENT

The final model was deployed to predict vitamin deficiency diseases using raw user input.

Deployment Process:

- Loaded the saved Random Forest model
- Loaded the saved feature selector, scaler, and column structure
- Collected raw user inputs such as health details, symptoms, lifestyle habits, and diet type
- Applied the same pre-processing steps used during training, including encoding and column alignment
- Performed feature selection and scaling
- Predicted the final disease class

Output:

The system predicts one of the following classes:

- Anemia
- Healthy
- Night Blindness
- Rickets/Osteomalacia
- Scurvy

This deployment ensures that new user data follows the same pipeline as the training data, providing consistent and accurate predictions.

12. CONCLUSION

This project successfully developed a Machine Learning-based Supervised Classification system for predicting vitamin deficiency diseases using patient health, nutritional, and lifestyle data. Data pre-processing, exploratory data analysis, feature selection, and model comparison were performed to build an effective prediction system. Among all tested models, Random Forest provided the best performance and was selected as the final model.

The deployed system can take raw user input, process it through the trained pipeline, and accurately predict possible vitamin deficiency diseases. This project demonstrates how machine learning can support early disease prediction and improve nutritional health awareness.